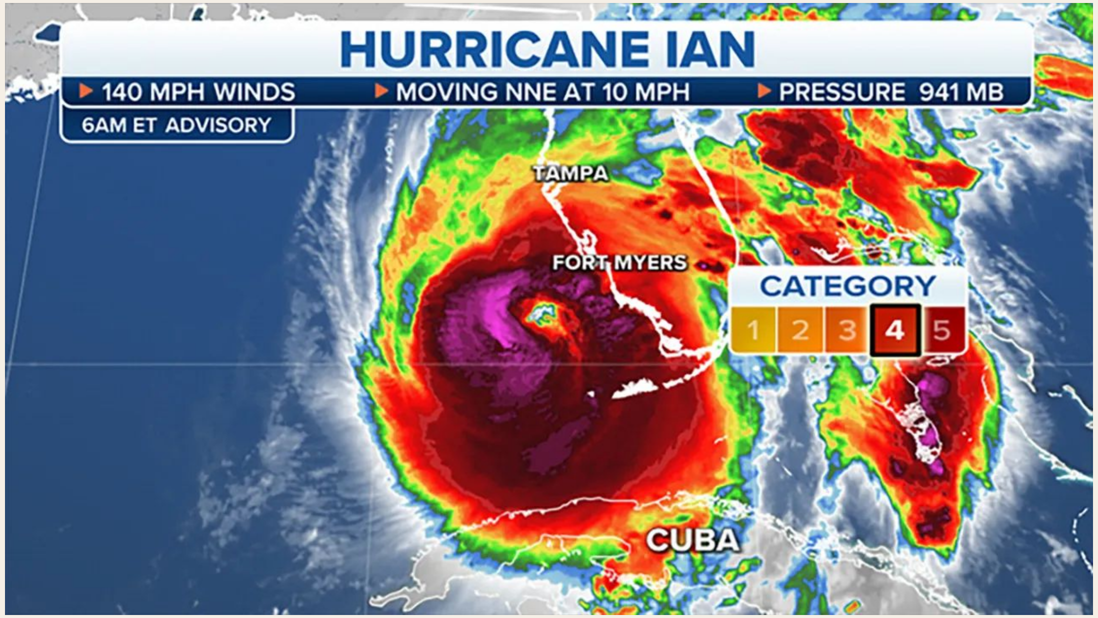
Hurricane Ian Building Damage Classification

Fort Myers Beach, Florida / September 2022

Mark Deng · Jun Luu · Sam Sen

University of Pennsylvania

BACKGROUND + THE PROBLEM



$113B

in estimated damages from Hurricane Ian (NOAA, 2023)

2,298

buildings classified in our Fort Myers Beach study area

Category 4

with 150mph winds

After Hurricane Ian made landfall on September 28, 2022, emergency managers needed to know which buildings were still standing, and where.

Traditional post-disaster surveys rely on field inspection. They are slow, labor-intensive, and often arrive after resources have already been committed.

WHO BENEFITS

Emergency response agencies, FEMA damage assessors, local building departments, and insurance adjusters working in the first 72 hours after a storm.

STUDY AREA

Fort Myers Beach barrier island, before and after

Maxar high-resolution satellite imagery

PRE IMAGE	POST IMAGE	RESOLUTION
Jan 2022	Sept 30, 2022	~0.5 m RGB

DATA

457 manually labeled buildings, three damage classes

Human-labeled building footprints, overlaid on post-Ian imagery

CLASS DISTRIBUTION

No damage 140

Damaged 202
Moderate Damage

Destroyed 115
Servere Damage

128x128 chips extracted per building · 70/15/15 train/val/test split

Per-building image chips were generated from the minimum enclosing square bbox, buffered for context, and resampled to 128×128

APPROACH

Four models, progressive complexity

We established a simple floor, tested the pretrained xBD baseline out of the box, then fine-tuned it in two stages.

MODEL 1

Simple CNN

Three conv blocks, trained from scratch on post-event chips. Sets a baseline floor.

BASELINE

MODEL 2

xBD baseline

Pretrained xBD weights applied directly, with 4-class output remapped to our 3-class scheme.

ZERO-SHOT CHECK

MODEL 3

Fine-tuned (post only)

xBD backbone frozen, shallow CNN and dense head retrained on local labels.

POST-ONLY TRANSFER

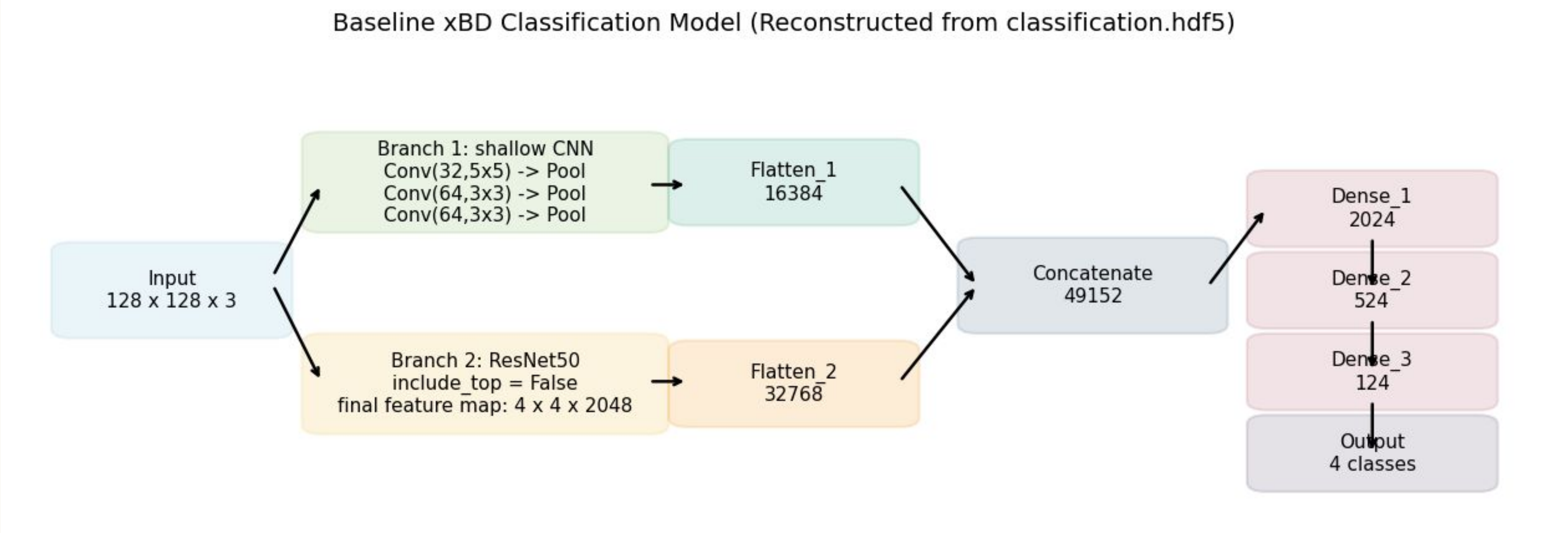
MODEL 4

Bi-temporal fine-tuning

Shared encoder on pre and post chips, fused as [f_pre, f_post, |f_post - f_pre|].

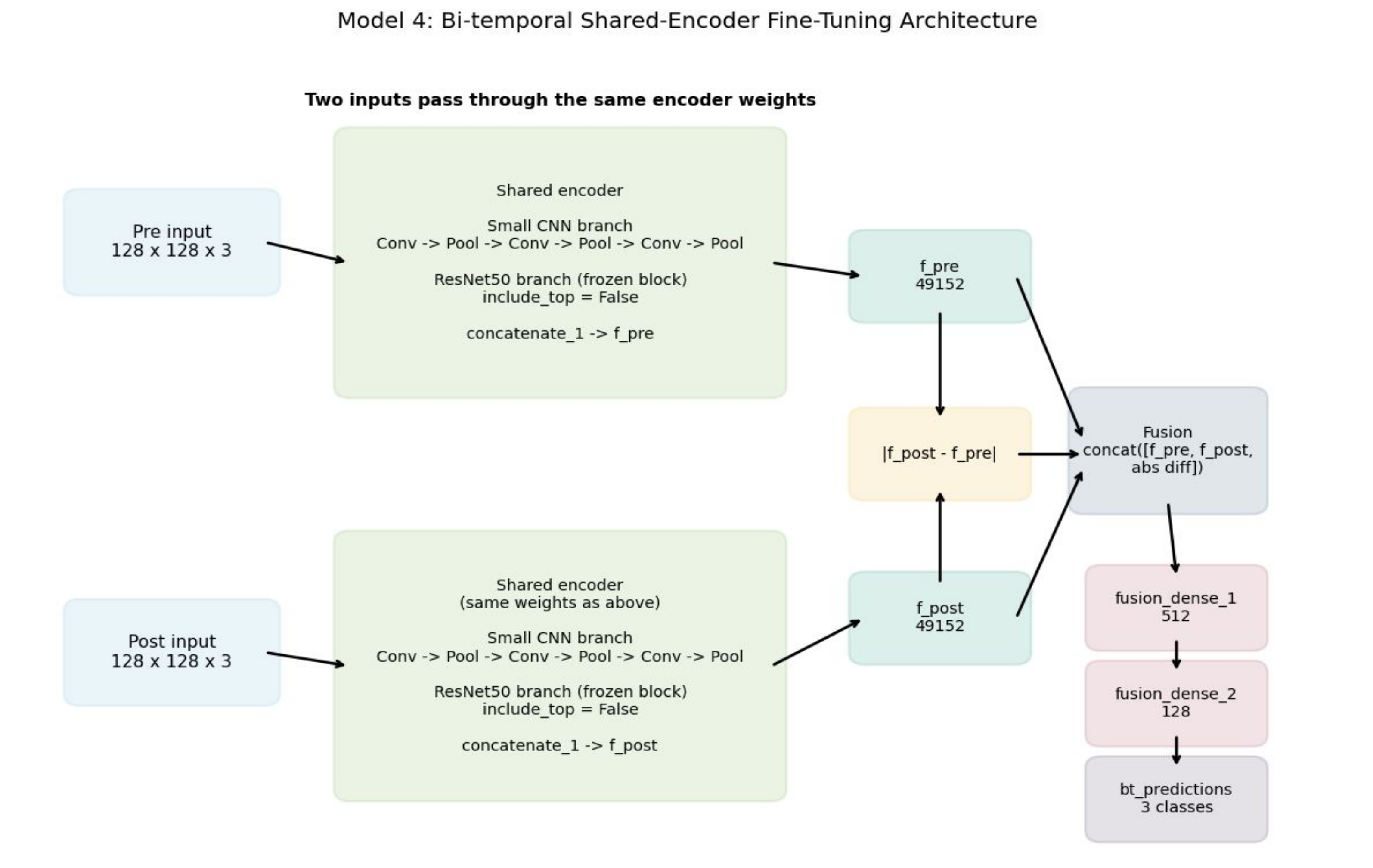
OUR MAIN MODEL

xBD Model Architecture



ResBet frozen, shallow CNN and dense head fine-tuned on local labels for Model 3.

Shared encoder with bi-temporal feature fusion



KEY DECISIONS

Shared weights

Pre and post chips pass through the same encoder, so the model compares features in a consistent representation space.

ResNet50 frozen

Deep backbone kept fixed to preserve pretrained disaster features and avoid overfitting to our 457 samples.

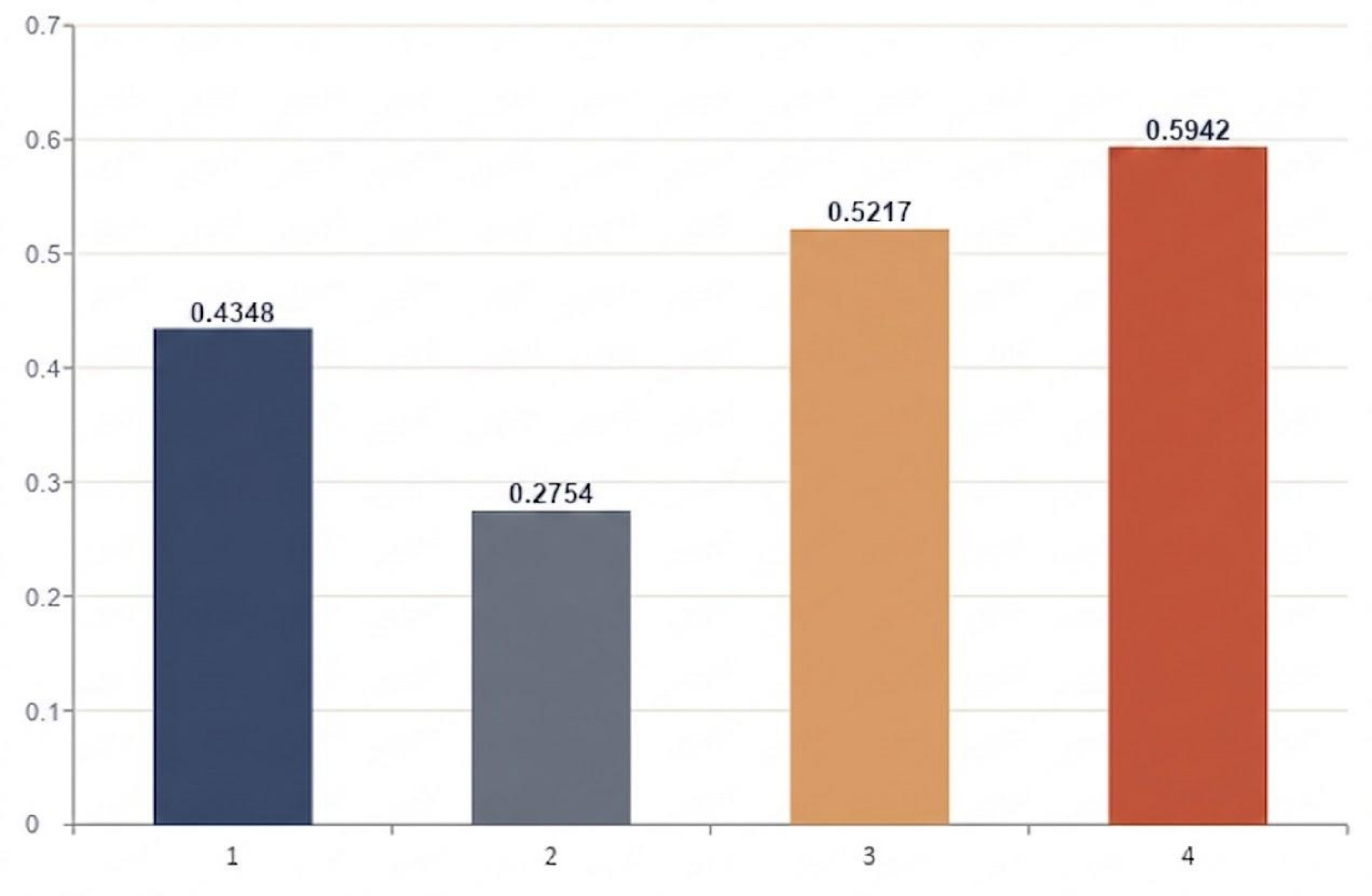
Shallow CNN trainable

Local texture layers (conv2d_1 to 3) adapt to Fort Myers roofs and debris patterns.

Fusion with difference

concat([f_pre, f_post, |f_post - f_pre|]) gives the head explicit change signal.

Model 4 Accuracy



Test accuracy on 69 held-out buildings · stratified 15% test split

BEST RESULT

59.42%

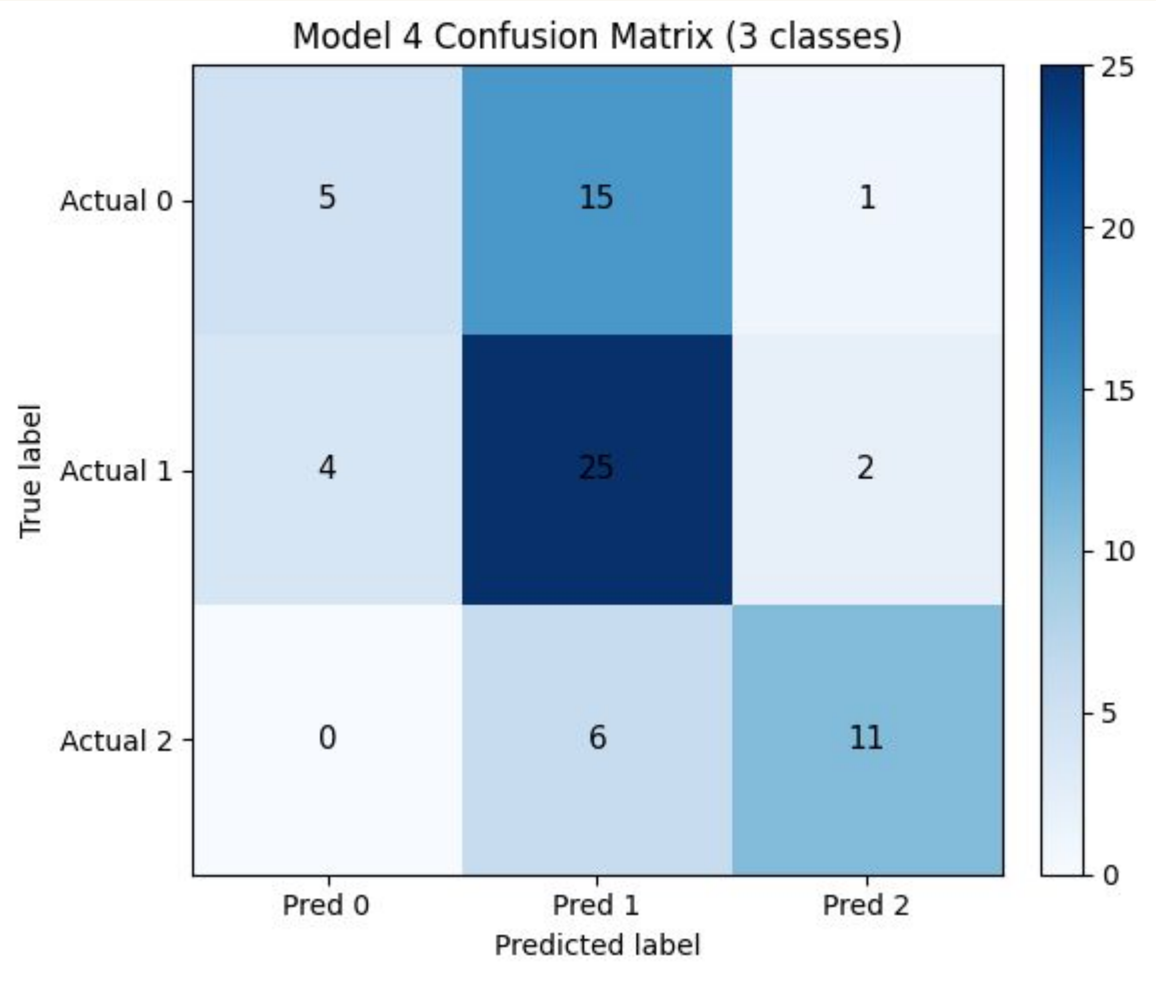
Model 4 test accuracy

READING THE NUMBERS

Zero-shot transfer fails. The pretrained xBD model collapses to "no damage" on Fort Myers, scoring below random.

Fine-tuning recovers performance. Model 3 alone beats the scratch CNN by 9 points. Adding the pre-event image gives another 6 points.

Diagnostics for Model 4

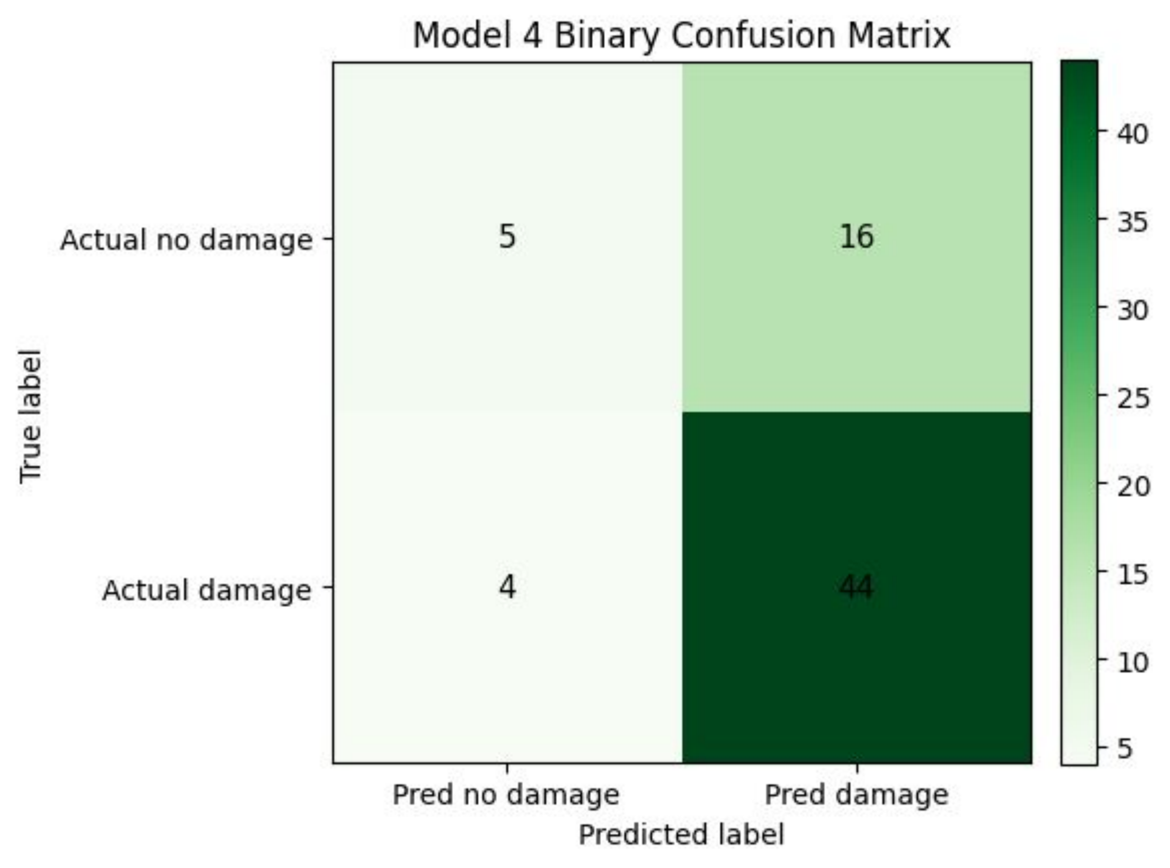


Model Performance Metrics: Recall Analysis by Label

Label	True Sample Count	Correctly Identified	Recall
0	21	5	0.238095
1	31	25	0.806452
2	17	11	0.647059

- **The bi-temporal model shows strong recall for damage-related labels.**
- **The main source of error is confusion between no-damage and moderate damage.**
- **The model is great at identify building damage.**

Model 4 as a Binary Damage Detector



Model 4 binary accuracy (no damage vs damage): 0.7101

So, the model is strong at detecting whether damage is present, but less precise when it comes to grading its severity.

DEPLOYMENT

Model 4 applied to all 2,298 buildings in the study area

PREDICTED COUNTS

No damage	363	16%
Damaged	1132	49%
Destroyed	803	35%

WHERE IT STRUGGLES

The model reliably separates damaged from undamaged. It struggles to determine between class 0 from class 1.

Limitation, Discussion, and Policy Implication

Small labeled set

457 buildings in one disaster limits what the model can generalize to. 57.97% accuracy should be read with this scale in mind.

- The bi-temporal model achieved the best overall performance, although its improvement over the post-only fine-tuned model was modest, likely due in part to the small test set. More importantly, both fine-tuned models clearly outperformed the two non-adapted baselines, showing that transfer learning with a small amount of local labeled data can effectively adapt a pre-trained damage model to a new disaster context. This suggests a practical workflow for smaller disaster-affected cities that need a rapid building-level damage assessment pipeline but do not have the resources to build a large training dataset from scratch.

RGB only

We did not use SAR, LiDAR, or building attributes. These could disambiguate partial damage from shadow and debris.

- From an modeling perspective, the bi-temporal model is especially promising for large-area deployment because it is likely to be more robust in full-area prediction for its dual input. It also showed strong performance in identifying damaged buildings, with high recall for damage-related classes, and performed well when simplified to a binary no-damage versus damage task. This means the model could support not only multi-class damage mapping, but also fast screening of affected buildings to help prioritize post-disaster response and recovery.

Thank you. Questions?